Chapter 3: Minimum potential energy

Solutions to the exercises

Exercise 4: Stress, the force on a surface, and a dam

1. The force per unit area on a surface with the unit normal n is σn.

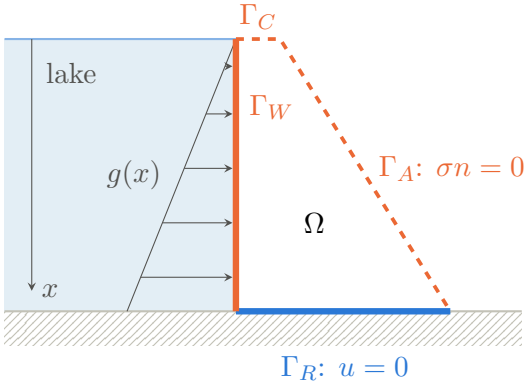
a) For $\sigma = \begin{pmatrix} s & 0 \\ 0 & 0 \end{pmatrix}$: $n = (1,0)$ gives $\sigma n = (s,0)$, the cross section is pulled along its normal with s. $n = (0,1)$ gives $\sigma n = 0$, the lateral faces are free. $n = (1,1)/\sqrt{2}$ gives $\sigma n = (s,0)/\sqrt{2}$, which is not parallel to n: the part along n is $\sigma n \cdot n = s/2$, the part within the surface has the size $s/2$, a shear force.

b) $\sigma n = -p\,n$ for every n: the water pushes against every surface along its normal, with the same strength in all directions.

2. Linear elasticity has $O(A) = 2$, $s = 1$, $m = 0$: the displacement ($c = 0$) is an essential, the traction σn (first derivatives, $c = 1$) a natural boundary condition.

- Γ_R (rock bed): **Dirichlet**, the dam is fixed to the rock: $u = 0$.
- Γ_W (water side): **Neumann**, the water pressure acts against the outward normal and pushes the dam downstream (item 1 b with $p = g(x)$): $\sigma n = T = -g(x)\,n$, with $n = (-1,0)$ on the vertical face.
- Γ_C (crest) and Γ_A (air side): **homogeneous Neumann**, free surfaces: $\sigma n = 0$.

In $g(x) = \rho\,g\,x$, x is the depth below the water surface: with the vertical coordinate y and the water surface at $y = H$, $x = H - y$.



3. The test functions vanish where u is prescribed, $v = 0$ on Γ_R; there the traction (the unknown reaction force of the rock) drops out of the boundary integral. On Γ_W the traction is $T = -g(x)\,n$, on Γ_C and Γ_A it is zero:

$$\int_{\partial\Omega} T \cdot v \; ds = -\int_{\Gamma_W} g(x)\,n \cdot v \; ds = \int_{\Gamma_W} g(x)\,v_1 \; ds.$$

The variational form of the dam: find u with $u = 0$ on Γ_R such that

$$\int_\Omega \sigma(u) : \varepsilon(v) \; dx = \int_\Omega f \cdot v \; dx + \int_{\Gamma_W} g(x)\,v_1 \; ds \qquad \text{for all } v \text{ with } v = 0 \text{ on } \Gamma_R,$$

with the own weight of the concrete $f = (0, -\rho_c\, g)$ as body force ($\rho_c \approx 2400\,\mathrm{kg/m^3}$, water $\rho = 1000\,\mathrm{kg/m^3}$). The essential condition on Γ_R is built into the function space (`DirichletBC`), the inhomogeneous natural condition on Γ_W appears as a boundary integral, and the homogeneous ones on Γ_C and Γ_A by omission.

Strong form for comparison: $-\nabla \cdot \sigma(u) = f$ in Ω, $u = 0$ on Γ_R, $\sigma n = -g(x)\, n$ on Γ_W, $\sigma n = 0$ on $\Gamma_C \cup \Gamma_A$.